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Financing New Ventures

Overview

Entrepreneurs launching new ventures face a wide spectrum of financing choices. Each startup venture will require its own particular financing strategy, but common questions repeatedly surface about the many financing options that entrepreneurs see around them (e.g., banks, venture capital, angels). This note provides a framework to help think about and organize these financing choices by relating the firm's financing strategy to the firm's business model and product strategy.

This note helps entrepreneurs think about:

- **Funding Sources** – From whom should you raise money? Which sources of financing are most appropriate for your new venture? How does the choice of the initial investor affect who the later investors will be?
- **Pitching Strategies** - What are funders looking for and how should you structure your business model to maximize your chances of raising money from them? How do you “pitch right for your type”?
- **Financing Tactics** - How much money should a venture raise? How is valuation determined? What other aspects besides valuation are important to consider?

We start by reviewing the basic motivations of two main types of investors: debt and equity investors. This background, and particularly how investor types view risk and uncertainty, allows us to build a mapping of financing options. This mapping will then be used to evaluate the more tactical questions identified above.

Debt versus Equity Capital

Debt investors (e.g., typical bank loans) lend a fixed sum of money for a specified period of time at a given interest rate. An inherent aspect of the debt contract is the fact that in the best case scenarios, investors' upside is bounded—that is, even if the entrepreneur is extremely successful, the debt investor only receives a return equal to their principal plus the pre-specified interest. In fact, as long

Professors William R. Kerr and Ramana Nanda prepared this note with the assistance of Senior Researcher Liz Kind as the basis for class discussion.

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as the value of the company is greater than or equal to the principal plus interest owed to the bank, the debt investor receives the same payout.

While the upside is limited, debt investors can still lose the full value of their principal in the event of a default. As a result, debt investors are primarily concerned about downside protection and only lend to less risky ventures, where the technology or business model is proven and where there is some way to collateralize the assets of the firm. In addition, debt investors seek businesses with steady, predictable cash flows where interest payments can be covered with income. Debt investors tend to press entrepreneurs to take on less risky strategies and business models, since the investors do not benefit from the increased risk, but do face the prospects of higher losses from a more risky strategy. **Appendix A** provides additional details on the different forms of debt financing available to entrepreneurs and the requirements that debt investors often stipulate in order to minimize the downside risk of their investments.

Many startup ventures alternatively rely on equity capital like venture capital (VC) and angel investments.¹ Equity investors receive a long-term ownership claim over the venture. Equity investors can also lose the full value of their principal, but the key difference relative to debt investors is that their upside is unbounded. That is, the return to the equity investor is proportionate to the value of the firm they invest in. If the firm becomes worth billions of dollars, their return is commensurately large. This upside participation allows them to take on much more risky and uncertain projects than debt investors.

The following example helps to show this in greater detail: an investor is looking to invest \$10,000 and has the choice of two projects. One is a “safe” investment that is certain to return \$15,000. The second is a “risky” investment that has a 10% probability of returning \$500,000 and 90% of failing completely. The expected value (probability of success $\times$ value if successful) of the first project is \$15,000 ($100\% \times 15,000$) and the expected value of the second project is \$50,000 ($10\% \times 500,000 + 90\% \times 0$).

Despite the expected value of the second project being more than three times that of the first, a debt investor seeking a 10% return on investment will only lend to the first project. This is because the most that a debt investor can realize from either project is \$11,000. Hence the debt investor will not benefit from the higher return the second project can generate if it is successful, while he still faces a 90% chance that the project will fail, in which case he will not even receive a penny. On the other hand, a (risk neutral) equity investor will choose to invest in the project with the highest expected value, as the equity investor benefits from the higher returns the project can generate if it is successful, even if there is much greater chance of failure.

Companies that face substantial uncertainty therefore rely on equity investors for external finance. While equity investors are willing to bear more risk, this comes at a price. The interest rate on debt investments tends to fall in the 10-15% range, while the hurdle rate of return that VCs and angel investors factor into their models when assigning a valuation on a startup can be as high as 50-75%.² Aside from the higher “price” that equity investors charge in return for bearing more risk, equity investors also try to manage this risk by investing in sectors where they have expertise and by influencing management decisions through their role on the board. These mechanisms increase the probability of success and thereby allow equity investors to also increase the expected value of the

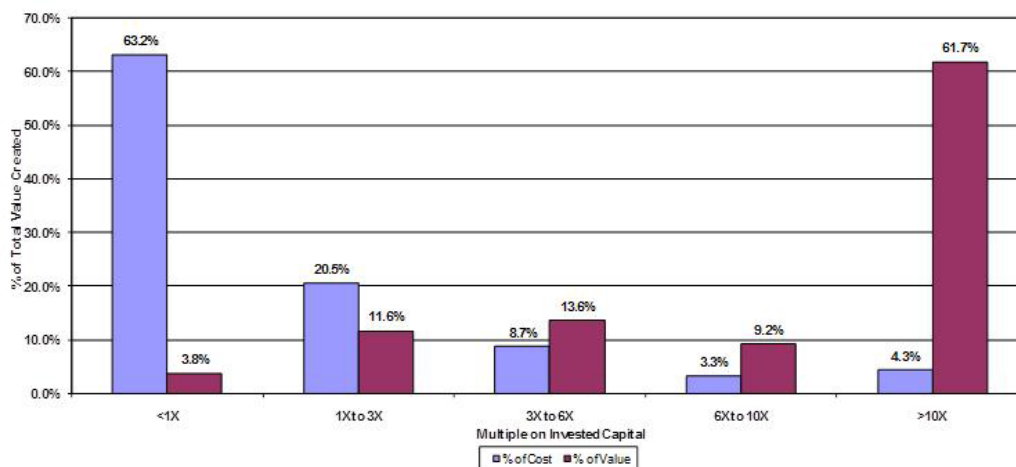
¹ As rough definitions, angel investors are individuals investing their own money in new ventures. VC investors are professionally-managed firms that invest in new ventures using funds developed from limited partners like pension funds, endowments, and wealthy investments.

² **Appendix B** provides a brief overview of the intuition behind VC valuations.

project through their own involvement.³ Monitoring and governance are also more important for the equity investors, given that their investments are often not collateralized and hence they stand to lose more in the event that the startup fails.⁴ Finally, equity investors often seek a portfolio of investments in order to diversify the risks they face. As long as their investments face idiosyncratic and uncorrelated risks (e.g., the technology risk of one venture is unrelated to the same risks in another portfolio company), equity investors can make significant returns even with very risky investments.

Figure 1 documents how even a very risky set of investments can lead to substantial returns, using data from a prominent VC's portfolio of investments. As can be seen from the figure, 60% of the dollars invested by the VC were for projects that ultimately failed and were liquidated below the cost of the investment. These projects contributed to less than 4% of the returns of the portfolio. On the other hand, 8% of the dollars invested by this VC firm accounted for over 70% of the overall returns of the portfolio.⁵ It is important to note that at the time of the initial investment, all of these investments had the potential to realize exceptional returns. Indeed, given the high chance of failure, VCs will not invest in projects that do not have the potential to be "winners." The VCs just do not know which of their investments will end up as winners and which ones will end up being failures. Put differently, the return distribution for VCs will end up looking like that in **Figure 1**, even though at the time of the investment, they believed that each startup might end up being a winner.

Figure 1 Breakdown of a Prominent VC's Portfolio



Source: Analysis by Professor William Sahlman based on 468 investments over the period 1990- 2006.

³ Note that this is also why entrepreneurs are often advised to take "smart money" for their startups. Experienced equity investors can help to increase the probability of success, as well as the value of the company if successful. Both these facets help increase the value of the company. We address these issues later in this note.

⁴ Even with collateral, equity investors have a lower likelihood of collecting any residual value due to their subordinate position in the capital structure. However, VC and angel investors also try to protect their downside using complex contractual features such as convertible preferred stock that act more like debt in "bad states of the world". These details are outlined in **Appendix C**.

⁵ Data on VC investments tend to be private so that a large-scale analysis is difficult to document. Moreover funds with investments primarily in software have somewhat different return profiles than those in more capital-intensive sectors such as biotechnology. However, discussions with investors confirm that skewed distributions highlighted in Figure 1 are a common feature in venture portfolios, even for the best venture funds.

An important facet of the VC investment process therefore involves putting a large enough number of investments in a portfolio to increase the chances of funding a “home run.” Moreover, since VCs rarely make money from more than half of their investments, they need to own a sufficient share of the successful firms at exit in order to generate adequate dollar returns to more than cover the failures. As a consequence, any individual investment cannot be too capital-intensive relative to the size of the overall portfolio, so that equity investors can both diversify across investments and have a sufficient fraction of the overall equity in the successful investments to cover the losses they incur in over half of their portfolio. **Figure 2** highlights this trade-off using an example of a pro-forma that VCs show to their limited partners (that is, the investors in their funds).⁶

On a similar note, it is important to note that the VC economics do not mean that a VC necessarily becomes more successful by lowering the number of failures in its portfolio, although the less investment that one makes in a failing company is clearly a positive. The nature of the portfolio is that one job of a VC is to close underperforming investments in favor of shifting resources to higher-return opportunities, moving invested capital from the left to right in Figure 1. Some of the most successful VC firms have among the industry’s highest failure percentages because they are better at managing this process and extracting higher returns from their portfolio. Better identification and management of failures in turn allows the VC to take on a higher risk portfolio and achieve greater returns.⁷

Figure 2 Example of Pro-Forma Shown by VCs to their Investors

Category of Outcome	Projected Value at Exit	Dollar invested per company	Share owned at Exit	Expected # investments	Total \$ invested	Total \$ Return
Early Failure	-	\$ 5 M	n/a	5	\$ 25 M	0
Complete write off	-	\$ 8-15 M	n/a	5	\$ 55 M	0
Money back	\$ 50 M	\$ 8-15 M	20%	5	\$ 55 M	\$ 50 M
Successful exit (low)	\$ 200 M	\$ 8-15 M	20%	5	\$ 55 M	\$ 200 M
Successful exit (medium)	\$ 350 M	\$ 8-15 M	20%	5	\$ 55 M	\$ 350 M
Successful exit (high)	\$ 500 M	\$ 8-15 M	20%	5	\$ 55 M	\$ 500 M
Total					\$ 300 M	\$ 1,100 M

Source: Shikhar Ghosh and Ramana Nanda, “Venture Capital Investment in the Clean Energy Sector,” Harvard Business School Working Paper 11-020 (2011).

A Mapping of Funding Sources

The debt versus equity discussion highlights that different types of investors seek very different types of startup firms as investment opportunities. Debt investors avoid risky and uncertain projects; equity investors invest in projects with high uncertainty, and moreover equity investors may actively *avoid* investing in less risky projects due to their lower returns. These differences in investor incentives can influence the product market strategy of funded startups in very different directions.

⁶ Details of the structure of VC partnerships and their relationships with Limited Partners are outlined in **Appendix D**.

⁷ Ramana Nanda and Matt Rhodes-Kropf, “The Financial Guillotine,” HBS Working Paper (2011).

Debt investors often advocate strategies with more stable returns, while equity investors may push for strategies that increase the upside returns. Understanding these different interests on the part of investors is a key aspect of a successful financing strategy for any entrepreneur, as it will have a bearing on the trajectory of the startup.

Our mapping in **Figure 3** focuses on two key dimensions to consider when raising money from investors: capital requirements and the uncertainty of the venture's opportunity. New businesses fall into different areas in this two-dimensional space, and each quadrant lends itself to different financing strategies.

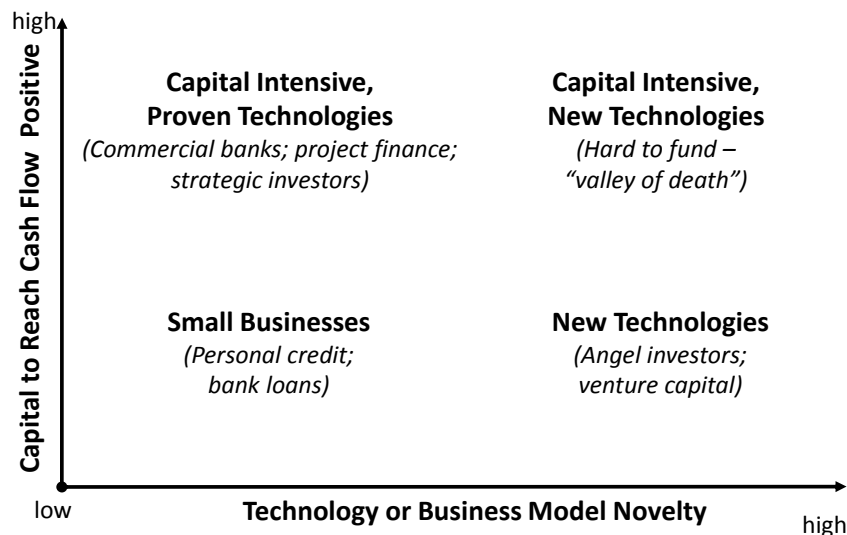
Startups in the left quadrants are less risky and can be financed through debt investments, which tend to be a less expensive source of external finance for entrepreneurs. Contrary to what some might expect, the vast majority of new firms fall in the **bottom left quadrant**. These startups have modest capital requirements and proven business models (e.g., restaurants, auto parts dealerships). To give a sense of the size of this quadrant, 20 million of the 26 million firms in the U.S. are owned by self-employed individuals, full or part time, *without any paid employees*. Of the remaining six million businesses, 80% have 20 employees or fewer. Even among high-growth firms such as those in the Inc 500, business models are often quite similar to those of the founders' prior businesses (before becoming entrepreneurs). These proven business models facilitate funding through personal finance, credit card debt, bank debt (either in a business or personal loan), and internally-generated funding.⁸

More capital-intensive projects, such as those within existing firms, may rely on commercial banks for debt finance. Debt finance is also the major source of capital for entrepreneurs looking to buy and turn around an existing business.⁹ Still more capital-intensive projects, such as coal power plants, infrastructure developments, and utility-scale solar or wind farms with well-developed technologies, fall in the **upper left quadrant**. While these ventures may need to raise hundreds of millions of dollars, the minimal uncertainty in terms of the technology and the stable, predictable, future cash flows easily afford debt instruments. These projects regularly receive project finance and/or structured finance investments from large commercial, investment banks, or strategic investors.

⁸ For example, Amar Bhidé, "On the Origin and Evolution of New Businesses," (Oxford: Oxford University Press, 2000). The Kauffman Foundation Entrepreneurial Survey provides detailed information on financing for representative startups annually. In 2004, the average financing for a new venture in the US was \$40,171 (51%) from the firm's owner and internal sources, \$6,979 (9%) from outsider equity, and \$31,255 (40%) from outsider debt.

⁹ For example, "Jim Sharpe: Extrusion Technology, Inc. (Abridged)," HBS Case No. 603-084 (Boston: Harvard Business School Publishing, 2003), by H. Kent Bowen and Barbara Feinberg.

Figure 3 Entrepreneurial Finance Framework



Source: Case authors. An academic discussion of this framework is William Kerr and Ramana Nanda, “Financing Constraints and Entrepreneurship,” in *Handbook on Research on Innovation and Entrepreneurship*, edited by David Audretsch, Oliver Falck and Stephan Heblich (Cheltenham, UK: Edward Elgar Publishing, 2011).

The right-hand side of the chart represents businesses with higher levels of innovation and uncertain outcomes. The more novel and unverifiable the technology proposed by the entrepreneur, the more difficult it is for traditional financial institutions to evaluate the creditworthiness of the project at hand. Many such startups are likely to have fewer tangible assets with verifiable valuations that can be pledged for a bank loan. Moreover, some of these projects have business models that require them to grow aggressively in order to be successful (e.g., two-sided platforms or “winner take all” markets) so that they cannot rely only on internally generated cash flows in order to finance their aggressive growth. These projects tend to be too risky to attract the cheaper debt finance, and hence require equity investors to step in and finance their growth and development.

Businesses in the **bottom right quadrant** combine relatively modest capital requirements and high uncertainty (e.g., software startups, medical device startups). These opportunities tend to attract angel and VC investors as their relatively modest capital requirements allow VC investors to make significant investments in these firms while still preserving diversification. Although these types of startups (such as Google and Facebook) can be high profile, they are rare and constitute less than 1% of the new firms that are founded each year in the U.S. Typical estimates suggest that both angel and VC markets in 2010 are about \$20 billion in size, although estimates of the angel market are very difficult to make given the private nature of the investments.

As outlined in **Figures 1 and 2**, angel and VC investors can make returns even as individual deals have irreducible risk. This portfolio logic also highlights an important limitation to how much capital investment these firms can make in any one venture. The typical VC investor avoids investing too much in a single project in order to maintain diversification in their portfolio and a sufficient share of the successful firms at exit. VC investors typically fund investments in the range of \$1 million to \$10

million. Angel investors often deploy less capital (often \$25,000 to \$50,000) and focus on earlier stages of the firm's lifecycle.¹⁰

Companies in the **upper right quadrant** are both extremely capital-intensive and based on novel technologies (e.g., supply-side clean energy startups such as thin-film solar or advanced biofuels startups).¹¹ These startups find it hard to attract capital and are said to fall in the proverbial "valley of death".¹² On one hand, they are too capital-intensive for traditional VC investors—the risk capital and coordination costs involved in funding a \$250 million demonstration bio-refinery may be too great for VC investors with \$300 million funds, even if the project has a positive net present value in expectation. Yet this quadrant is also too risky for project finance. While project finance firms are very comfortable funding highways, dams, coal-powered plants, and other well proven technologies that are equally if not more capital-intensive, they are reluctant to fund projects with unproven technologies that are much more likely to fail. As a result, the businesses in this quadrant often require government support and funding; strategic financing from large corporations is also possible in some industries.

Funding Sources and Business Models

What implications does this basic mapping have for how entrepreneurs run their businesses? While describing all of the potential connections between business models and financing options is beyond the scope of this note, we highlight here several broad connections, often in the form of a short case study.

First, these differences in financing options are influential in location decisions for startups. In most advanced economies, startups can obtain debt forms of financing readily. Sophisticated forms of equity financing, however, have very different degrees of penetration across advanced economies and their sub-regions. Moreover, these investors tend to look for local opportunities (e.g., a one hour drive or short plane flight) due to the extensive time required to evaluate, assist, and monitor the startup. These spatial patterns in funding types influence and reflect differences in the kinds of entrepreneurship and types of industries that emerge in different regions. An entrepreneur envisioning a new and novel business model in a location without these types of financiers is stacking the deck against the venture in terms of financing strategies.

Second, some ventures can be cleverly designed to reduce the capital needs or the uncertainty faced by financiers. This can be particularly valuable if the venture needs to avoid the "valley of death" and obtain funding from more traditional sources. As one example, Skyhook Wireless ("Skyhook") provides location awareness to portable electronic devices by observing the WiFi signals

¹⁰ These investors often syndicate the investment deals, where multiple VCs and angels invest into a startup at the same time. This syndication allows for greater investment amounts while keeping the exposure for each individual firm more manageable. Syndication can also bring additional benefits by pooling the resources of multiple investors (e.g., different professional networks) if the interests of the investors remain aligned.

¹¹ Ramana Nanda and Toby E. Stuart, "KiOR: Catalyzing Clean Energy." HBS Case No. 809-092 (Boston: Harvard Business School Publishing, 2009).

¹² Note that some industry participants also refer to the lack of seed and early stage finance for startups as the "Valley of Death". This lack of early stage finance is a challenge for *all* startups, not just for very capital-intensive ones. In this note, our reference to the "Valley of Death" is very specific to projects that are too expensive for VC and too risky for project finance. See for example, Michael J. Roberts, Joseph B. Lassiter, and Ramana Nanda, "U.S. Department of Energy & Recovery Act Funding: Bridging the "Valley of Death", HBS Case No. 810-144 (Boston: Harvard Business School Publishing, 2010).

that surround the device and their relative strength.¹³ An essential ingredient for Skyhook's product is a mapping of the geographical location of all WiFi access points in a region. The founders could have concluded that they should raise \$100 million dollars or more in order to map out the US, pushing Skyhook into the "valley of death" in terms of financing. This price tag is in fact what many competitors and customers believed Skyhook paid.

Skyhook, however, was able to develop its network having raised less than \$17 million dollars from investors. First, they experimented and proved the concept sufficiently to raise early financing from angels and VC investors. Skyhook next developed inexpensive equipment and algorithms that could map WiFi beacons from a moving car without requiring driver intervention. Finally, they hired their drivers from Craigslist.org at \$15 per hour with no mileage allowance. Very quickly, Skyhook developed a workable network that allowed the company to launch early operations, using subsequent data flow from its devices and retained earnings to further enhance the quality. Through smart and lean management, Skyhook developed a very valuable asset at a fraction of the typical cost and (quite importantly) within the budget of standard VC and angel investors.

Third, changes in the risk and uncertainty profile of a project can jeopardize its financing. Entrepreneurs often fail to see this connection when contemplating business model changes, simply assuming instead that all of the financiers will go along. TA Energy was founded as power generation facility in Turkey in 1998.¹⁴ The project combined global power generation partners with local customers in Turkey. The project developers secured over \$30 million dollars in project financing for the proposed facility from international banks. The project almost unraveled, however, at the last minute when the anchor tenant of the facility wanted to substitute its own construction arm for the international contractor previously selected. While the local construction company had deep experience in Turkey, it had never developed a power generation facility before. The international contractor, however, had extensive experience in building these types of plants.

For the international banks and partners, the uncertainty introduced by bringing in an inexperienced local contractor was too much for the project financing model they had established. It is important to note that whether or not the local construction firm would in fact do a good job was less important for financing purposes than the expectations of investors beforehand. The investors balked at providing the extensive debt financing if the venture became more uncertain. In effect, the potential change in contractor was dragging the opportunity to the right towards the "valley of death." The project ultimately went forward by keeping the international contractor (and thus the original capital-uncertainty profile) and finding other ways to satisfy the anchor tenant.

Fourth, it is important for entrepreneurs to understand that past financing decisions, especially with respect to equity investors, have lasting implications on subsequent business model decisions. Sirtris Pharmaceuticals ("Sirtris") confronted this challenge as a young biotech company doing very promising research into age-related diseases.¹⁵ Sirtris had established a link between resveratrol, a compound found in grapes that produce red wine, and sirtuins, a family of enzymes with links to improved longevity and health. The core of Sirtris focused on developing sirtuin-activating drugs that would be useful for diabetes and similar conditions. Characteristic of drug development, Sirtris' research had a very small probability of successfully navigating all of the steps to final approval of

¹³ Shikhar Ghosh and Thomas R. Eisenmann, "Skyhook Wireless," HBS Case No. 809-119 (Boston: Harvard Business School Publishing, 2009).

¹⁴ William R. Kerr, Daniel Isenberg, and Ant Bozkaya, "TA Energy (Turkey): A Bundle of International Partnerships," HBS Case No. 807-175 (Boston: Harvard Business School Publishing, 2007).

¹⁵ Toby Stuart and David Kiron, "Sirtris Pharmaceuticals: Living Healthier, Longer," HBS Case No. 808-112 (Boston: Harvard Business School Publishing, 2008).

the Food and Drug Administration (e.g., due to side effects). However, they offered blockbuster potential if successful.

Sirtris management debated a second route of selling their proprietary formulation of resveratrol through nutraceutical markets. This route was attractive as nutraceuticals was mostly unregulated, allowing Sirtris to enter quickly, and the company was benefitting from substantial media attention. There was also no doubt that money could be made. Yet, Sirtris' long-term investors had substantial concerns that this side effort would distract from or even tarnish Sirtris' pursuit of the blockbuster opportunity. In the context of the investors' portfolios of ventures, the investors wanted Sirtris seeking only the most spectacular opportunities possible. Given the balance of risk and reward that the investors were trying to achieve, they wanted Sirtris to "swing for the fences".

A related phenomena also occurs when a startup's financing mixes different types of investors. E-Ink provides an exaggerated example of this phenomenon, which exists across all sizes of venture investing (e.g., mixing unsophisticated angel and VC investors).¹⁶ E-Ink was founded in 1997 to develop a MIT-based technology oriented towards electronic paper; a later version of its technology would be incorporated in products ranging from low cost cellular phones to the Amazon Kindle. Due to the great interest in the many paths its technology could take, E-Ink's early investors included VC firms, a diversified newspaper and communications company, a large wireless equipment maker, and others. This diversity of investor backgrounds, goals, and time frames created competing interests over E-Ink's future direction once it became clear that E-Ink would not be able to conquer as many markets in as quick of a timeframe as initially thought.

Finally, most businesses remain in a single quadrant in Figure 3 for the entire lifecycle of the company, and this static portrait can help understand the needs of investors in that quadrant. Other ventures move across quadrants (usually right to left) as the company matures from experimental technology to commercialization with a known product. Reducing uncertainty unlocks more financing options for entrepreneurs. Understanding these transitions and how the financing around the successful venture will adjust is important; in many cases, exit opportunities will be required for investors as the venture moves out of their optimal financing environment (e.g., VC exits through IPOs or acquisitions).

Tactical Strategies for Financing Ventures

Having surveyed the broad range of financing options, the traits of various financiers, and how they influence the entrepreneur's business model, we close with some tactical advice for entrepreneurs when thinking about venture financing:

- *"Pitch right for your type"* – Pitching is a vital part of raising external venture financing. The above case examples highlight that an entrepreneur's pitch should reflect the interests of the financier. If attempting to raise VC financing, it is most important to establish the tremendous upside potential to your business opportunity. VCs look for opportunities that can generate \$50 million or more in revenues within five to seven years. Describing safe side bets that the entrepreneur's business can also make does not aid the presentation, and perhaps even detracts from it. On the other hand, the opposite is true for bank lenders. Entrepreneurs can scare bank lenders away by pitching extremely high

¹⁶ William A. Sahlman, "E-Ink: Financing Growth," HBS Case No. 800-252 (Boston: Harvard Business School Publishing, 1999).

return, low probability opportunities rather than the stability of their business model and its potential to generate steady returns for interest and loan payments.¹⁷

- *Understand that valuation is in the context of a financing negotiation* – When developing a financing strategy and pitching to investors, entrepreneurs must recognize that there is no “true” value to their venture. A variety of tools have been developed to help think about valuation: pre- and post-money valuation using the VC valuation method, discounted cash flow analysis, adjusted present value analysis, market transaction comparables, and similar. Ultimately, however, valuation is a negotiated outcome and the result of the relative bargaining power of the counterparties. Valuation is set in a context of a market that matches entrepreneurial ideas with entrepreneurial finance. That market can be hot or cold, and that too affects the deal outcomes that are struck.¹⁸
- *Deal terms have value, and it is easier to negotiate these than the value itself* – Deal terms affect both incentives and implicit valuation. Sometimes it is easier to negotiate on certain deal terms rather than directly on valuation. Many of the deal terms only have meaning or importance in certain scenarios (e.g., a low-valuation exit, a down round of financing). Entrepreneurs often focus on the upside valuation of the venture and neglect these terms that have reasonably good chance of being relevant. Sophisticated financiers will often gladly trade a higher initial valuation for more advantageous terms with respect to control, down round protections, and similar. The National Venture Capital Association provides additional resources on term definitions and common practices (www.nvca.org).
- *Who you raise equity money from is extremely important* – Equity investors come in all shapes and sizes. Some equity investors can add great value to the company in non-financial terms, while others provide just cash. While it is true that the very top tier of investors charge a higher price for their investment (i.e., take a greater equity share for a given capital infusion),¹⁹ the performance boost is also evident. For example, almost all of the success of the VC industry is associated with the top quartile of VC funds.²⁰ Entrepreneurs need to closely consider the non-financial assistance that different investors provide. Entrepreneurs should also remember that who they raise money from today and the terms of the initial deal have a large bearing on their future financing options.
- *Raise appropriate amounts of money to reach the next milestones* – Entrepreneurs often raise money in multiple financing rounds. This can be good for the financier, as it allows them to collect more information about the venture’s prospects before investing deep amounts of capital. It can likewise be beneficial for the entrepreneur, as continued success improves the terms for later financing rounds—raising money in increments after

¹⁷ For a general overview on pitching business opportunities to equity investors, see “Pitching Business Opportunities,” HBS Case No. 811-086 (Boston: Harvard Business School Publishing, 2011), by Lynda Applegate and William R. Kerr.

¹⁸ Good resources on entrepreneurial valuation methods include “Note on Free Cash Flow Valuation Models,” HBS Case No. 288-023 (Boston: Harvard Business School Publishing, 2006), by William A. Sahlman; “A Method for Valuing High-Risk, Long-Term Investments,” HBS Case No. 288-006 (Boston: Harvard Business School Publishing, 2009), by William A. Sahlman; and “A Note on Pre-Money and Post-Money Valuation (A),” HBS Case No. 802-179 (Boston: Harvard Business School Publishing, 2002), by Linda A. Cyr. A good resource for negotiations is “Negotiation Analysis: An Introduction,” HBS Case No. 801-156 (Boston: Harvard Business School Publishing, 2000), by Michael Wheeler.

¹⁹ David Hsu, “What Do Entrepreneurs Pay for Venture Capital Affiliation?” *Journal of Finance* 59 (2004), 1805–44.

²⁰ Steve Kaplan and Antoinette Schoar, “Private Equity Performance: Returns, Persistence and Capital Flows,” *Journal of Finance* 60 (2005), 1791–1823. These findings are also evident in micro-data studies. See “The Consequences of Entrepreneurial Finance: Evidence from Angel Financings,” by William R. Kerr, Josh Lerner, and Antoinette Schoar (2011).

successful milestones are reached results in the entrepreneur retaining more ownership overall. One way to think about raising money is to see it as a way to gain time to run experiments to open or close options.

Thus, entrepreneurs need to raise sufficient money to reach the next milestone (and it always takes much more time and money than anticipated!), so they should be careful to not slice the financing too thinly. It is very painful to be close to the next milestone and yet run out of money. And stronger cash reserves provide negotiation power for follow-on financing rounds. But, entrepreneurs also need to make sure that they don't raise too much money and suffer unnecessary dilution. This trade-off has been often called the horserace between fear and greed.

It is sometimes surprising for entrepreneurs to also learn that an extremely high initial valuation may not be desired. Clearly a stronger initial valuation lowers the dilution that the entrepreneurs give up in the initial financing. The challenge, however, comes with respect to obtaining milestones. Down rounds—that is, financing rounds where the value of the company slips from the previous financing round—are very painful. Investors impose expectations on a company when they decide to invest, and the magnitudes of those expectations are closely linked to the amount of funding and valuations a company receives. Very lofty expectations with regards to revenue, user adoption, etc., make it difficult for a company to keep pace with milestones.

Part of the art of entrepreneurial finance is having a nice progression of up rounds that generate lots of desire to be a part of the venture. If entrepreneurs don't leave sufficient room for this valuation growth, good opportunities can be severely damaged as unrealistic early expectations make access to later financing rounds difficult.

- *Build feelings of desire and urgency in investors* – Entrepreneurs are often disappointed when investors express lots of interest in their venture but are then slow to proceed forward. Investors are often evaluating multiple opportunities and may delay funding a promising venture while they wait for additional information. To the extent that entrepreneurs can instill a sense of desire and urgency in the investor to be a part of the deal, the venture may get a shorter funding clock.

Two great ways to create this urgency are to have alternative financing offers available or to have a great media buzz. These techniques are not always possible and can suffer from a chicken-and-egg problem, but other tactical options are possible, too. For example, VC investors often ask entrepreneurs to tell them the other financiers to whom they are pitching. Entrepreneurs should think hard about name dropping every firm they have scheduled, as the investors all know each other and can use this information to collude against the entrepreneur when negotiating. Strong investment presentations can also create this desire. The more that the investors feel this deal might happen without them, the better for the entrepreneur.

- *Do not run out of cash!* – This seems simple, but it is a cardinal rule that entrepreneurs too often violate. Entrepreneurial finance is negotiation, and entrepreneurs and investors are not on the same team until the deal is finalized. Remembering that negotiations are underway is essential, but can often get lost as the parties establish a great rapport during their interactions. Entrepreneurs should be thinking about the position of their counterparties (e.g., where the financiers are in their funding cycles, whether they have many other investment options). Nothing can put an entrepreneur in a worse bargaining position than running out of cash!

- *Start early and build a relationship* – Entrepreneurs often delay meeting with investors until they are actively fundraising, believing that earlier contacts are a waste of time and resources. Earlier and more regular interactions, however, can build relationships and networks in advance of funding needs. Investors think carefully about each investment and put great emphasis on the entrepreneur and his/her management team. This confidence can take a weeks or months to develop, and earlier interactions can build this confidence and personal connections. This can be particularly valuable for new entrepreneurs.

Entrepreneurs can show investors in advance that they are making great progress, hitting their milestones, and executing on their strategy. Having brief "update" meetings with VCs that convey this information and allow the VC to learn more about an entrepreneur is invaluable. It allows an investor to evaluate how the product is evolving and/or how the market is receiving the product. Thus, when an entrepreneur is ready to actually raise money, it becomes a much easier calculation and a faster decision for a VC because a lot of due diligence has already been completed; moreover, a desire to be a part of the deal has been fostered.

- *Demonstrate the "Mo"* – Each investor uses different criteria to calculate whether they will make a bet on an entrepreneur. But one factor that almost all investors use is whether the entrepreneur has what it takes to attract, recruit, hire, and inspire a high-performing team. Convincing people with awesome track records in other companies to jump ship and join a startup is probably most difficult for first-time entrepreneurs, but never easy for anyone. But demonstrating this leadership is invaluable. It shows investors that an entrepreneur has what it takes—the "Mo"—to inspire others to their cause.
- *Do your homework!* – This again seems simple, but entrepreneurs too often jump into raising financing without doing sufficient homework or building an appropriate team. A fundamental asymmetry exists for many entrepreneurs, who are much less experienced at entrepreneurial finance than the investors with whom they are negotiating. Entrepreneurs can level this playing field by learning about typical deal terms for their type of venture, researching investors and past entrepreneurs' experiences with them (e.g., www.TheFunded.com), and assembling the support of knowledgeable counsel. A good lawyer with experience in entrepreneurial finance deals can be particularly valuable, providing advice and contacts and often working on a contingency fee for a completed financing event. Entrepreneurs should, of course, always be aware of broader incentives when working with outside parties (e.g., a lawyer wanting to close the deal under any terms to collect the contingency fee).

Conclusion

Many ventures, especially high growth opportunities, require financing. While a broad spectrum of financing options exist, one can map these options into regular patterns with respect to levels of capital investments, business and technology uncertainty, and similar. Entrepreneurs should understand how their business opportunity fits into the desired profiles of potential financing options. This is important for deciding which types of financing to take on, how to successfully pitch to those types of investors, and how to manage and lead the venture after the financing is secured.

APPENDIX A: FORMS OF DEBT LENDING²¹

Lenders will allow firms to borrow against their expected ability to generate the cash to repay the loan. **Cash flow financing** is most commonly available from commercial banks, but can also be obtained from savings and loan institutions, finance companies, and other institutional lenders (e.g., insurance companies, pension funds). Cash flow financing is generally riskier than asset-based financing, and as a result, banks will frequently attempt to reduce their risk through the use of covenants on the loan such as minimum cash balances and limits on the company's debt/equity ratio. These covenants place certain restrictions on a business if it wishes to maintain its credit with the bank and attempt to protect the lender from actions which would decrease the likelihood of the bank's getting its money back. Obviously, a healthy business with little debt and high cash flow will have an easier time borrowing money than a new venture.

Asset-based financing requires a company to pledge or give the lender a first lien on specific assets that have market value. Because cash flow financing usually requires an earnings history, far more new ventures are able to obtain asset backed financing. In the event of a default on the financing payments, the lender can repossess the asset. Asset-based financing is available from commercial banks and other financial institutions. Insurance companies, pension funds, and commercial finance companies provide mortgages and other forms of asset backed financing. The following types of asset-based financing are generally available:

- **Accounts Receivable:** Up to 85% or 95% of the accounts receivable from creditworthy customers can usually be financed. The bank will conduct a thorough investigation to determine which accounts are eligible for this kind of financing. In some industries, such as the fashion business, accounts receivable are often "factored." A factor buys approved receivables for a discount from their face value, but collects from the accounts.
- **Inventory:** Inventory is often financed if it consists of merchandise which could easily be sold. Typically, 50% or so of finished goods inventory can be financed.
- **Equipment:** Equipment can usually be financed for a period of three to ten years. One-half to 80% of the value of the equipment can be financed, depending on the "saleability" or liquidity of the assets. Leasing is also a form of equipment financing, where the company never takes ownership of the equipment, but rents it.
- **Real Estate:** Mortgage financing is usually readily available to finance a company's plant or buildings; 75% to 85% of the building is a typical figure that can be financed.
- **Personally Secured Loans:** A business can obtain large amounts of financing if one of its principals (or someone else) is willing to pledge a sufficient amount of assets to guarantee the loan.
- **Letter-of-Credit Financing:** A letter of credit is a bank guarantee which a company can obtain to enable it to purchase goods. A letter of credit functions almost like a credit card, allowing businesses to make commitments and purchases in other parts of the world where the company does not have relationships with local banks.
- **Government Secured Loans:** Certain government agencies will guarantee loans to small businesses to obtain financing where they could not obtain it on their own. The Small Business Administration, the Farmers Home Administration, and other government agencies will guarantee bank loans.

This asset-based financing builds only on assets that the bank anticipates being able to sell in the event that a business loan fails. This "saleability" is linked to the thickness of markets for the asset

²¹ This section is derived from "New Venture Financing," HBS Case No. 802-131 (Boston: Harvard Business School Publishing, 2006), by Howard H. Stevenson and Michael J. Roberts.

and the dependency of the asset's value on how well the business loan performs. One can borrow a higher share against real estate than inventory due to these features. It is also helpful to note what is not included. Many entrepreneurs are surprised that banks are very reluctant to lend against their patents. The challenge with assets like patents is that the value of the patent is likely very closely tied to the performance of the firm and its loan. A business may fail because its technology did not work out, in which case these patents have very limited value. Precisely because the value of the patent is tightly linked to the success of the firm, banks have a hard time lending against them.

APPENDIX B: VENTURE CAPITAL VALUATION²²

Given that startups neither have a long history of operational performance, nor typically have a predictable growth trajectory from which to develop forecasts to a “steady state,” valuation tools based on discounted cash flow tend to be less helpful in developing a sense of a startup’s valuation. Instead, one can calculate the implied value of the startup by seeing how much money an investor put into the startup in return for a given share of the equity.

In this sense, pre- and post-money valuations are calculated only as a result of a financing event. They are the result of real dollars being exchanged for real shares, rather than a value being calculated in a spreadsheet. They are implied valuations because they are not the result of bottom up estimates of the firm’s value using projections, but are instead the imputed price derived of the last financing event of the firm. This fact highlights the fact that there is no “true value” to a startup with very uncertain outcomes. Two similar firms may get very different valuations in hot and cold markets or based upon whether the firm has an experienced or inexperienced entrepreneur. A startup with top tier investors in the seed round may get a very different second round valuation as a similar startup with less prestigious investors even if the business plan and entrepreneurs are quite similar.

How do venture capitalists and angel investors arrive at an estimate of the underlying valuation before they make their investment? While they have experience that may give them an understanding of the valuation range for a startup based on the market opportunity, the quality of the team and the stage of the venture (that often proxies for the underlying risk), such investors will often also “back into” the valuation, based on their targeted hurdle rate on their investments.

For example, suppose an investor is considering a \$4 million investment and targets a 50% compounded annual return on his investment over five years. This implies that in order to achieve his expected return, the \$4 million investment must be worth $(\$4M \times 1.5^5) = \30 million at the end of year five. In addition, suppose that by using historical data on price/revenue multiples for comparable companies, in combination with the VC’s projection of the firm’s revenues in five years, the investor estimates that the firm will be worth \$150 million dollars when it exits in five years through an IPO or an acquisition. Then, in order for the investor to make the required \$30 million from that investment, he must own $\$30 \text{ million} / \$150 \text{ million} = 20\%$ of the company at exit.

For simplicity, assuming that there are no further rounds of investment, the investor needs to then take a 20% share of the equity in return for the \$4 million investment in order to satisfy his hurdle rate of return. If he goes ahead and makes the \$4 million investment in return for a 20% share of the company’s equity, then the full value of the company’s equity at the time of the investment is worth $(\$4 \text{ million} / 20\%) = \20 million. This means that the implied post-money valuation of the startup is \$20 million. The pre-money valuation is simply the post-money value less the investment, that is, $\$20 \text{ million} \text{ less } \$4 \text{ million} = \$16 \text{ million}$.

This example highlights the fact that the pre- and post-money valuations are directly linked to the investor’s target return on his investment (which can also be seen as the discount rate or the “interest rate”). If the investor was to raise his target rate of return to 75%, then he would need $(\$4M \times 1.75^5) = \66 million at the end of year five to justify the investment. Given the projected value of the firm in year five, this would imply that the investor would need to take $\$66 \text{ million} / \$150 \text{ million} = 44\%$ of the firm in return for the \$4 million investment. In this instance, the implied post-money value of the firm falls to just \$9 million $(\$4M / 44\%)$. On the other hand, if the investor had a target rate of return

²² This section is based on “A note on Pre- and Post-Money Valuation”, Harvard Business School Note 9-802-179 (Boston: Harvard Business School Publishing, 2002), by Linda A. Cyr.

of 25%, then he would need $(\$4\text{M} \times 1.25^5) = \12 million at the end of year five to justify the investment. This would require him to take only an 8% share of the startup and hence value on the firm at \$49 million.

In a sense, the negotiation over valuation of the firm is a negotiation about the hurdle rate of return that the investor is using, combined with the estimate of the firm's value at exit. If investors believe the firm is more risky, because its technology or business model or team is unproven, then they are likely to put a higher discount rate or hurdle rate of return to justify the investment. Similarly, if they forecast the markets to be hot at the time of exit or the company to be a "home run" or blockbuster, then they expect the exit multiples to be higher and hence will require a lower share of the firm to generate the same dollar return from their initial investment. Finally, if there is a lot of interest in a particular company, investors will bid up the valuation of the firm, by offering to take less equity from the founders in return for the same amount of investment as another firm. All of these factors highlight that valuation of early stage startups is extremely subjective and is the result of a negotiation between investors and entrepreneurs. In this sense, there is no such thing as "true value" – the implied value stems from the share of the company the entrepreneur has to give up in return for the investment. The less they give up, the higher the implied valuation.

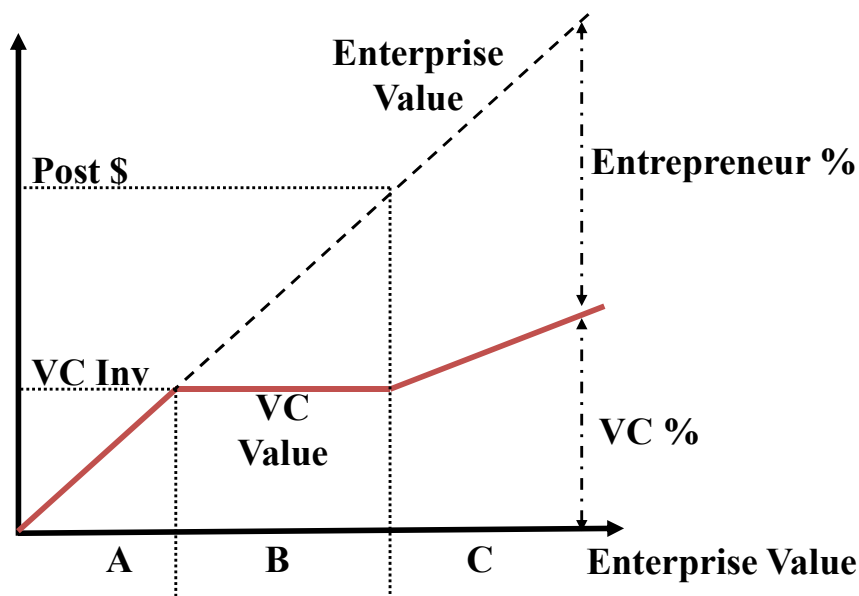
APPENDIX C: CONVERTIBLE PREFERRED STOCK

The discussion in the main text treats VC and angel investors as equity investors. In reality, their financing instruments often have a debt-equity conversion feature that is important for entrepreneurs to understand. Convertible preferred stock is preferred stock that can be converted from debt into equity at the shareholder's discretion. If the venture exits (e.g., is acquired) at low valuations, the shareholder does not convert into equity and thus gets the first proceeds typical of a debt holder. If the venture exits at high valuations, the shareholder converts into equity and obtains a portion of the venture's value.

To understand these features, it is helpful to define a hypothetical pre-money, post-money event building off of Appendix B. Take a scenario where an entrepreneur and VC strike a deal to invest \$4 million in a venture for a 40% stake. After the financing event, the implied "post-money" valuation of the deal is \$10 million (= \$4 million / 40% stake). This also implies that the "pre-money" valuation of the venture was \$6 million dollars (\$10 million - \$4 million, or 60% of \$10 million). We will also often say the entrepreneur owns 60% of the company after the financing event, and the VC owns 40%.

The ownership shares are correct, but they come with a big caveat. If the venture successfully exits soon after the financing event for \$100 million dollars, the VC will treat its investment as equity and obtain \$40 million; the entrepreneur will obtain \$60 million. However, at low exit valuations, the VC will not convert its debt into equity, and thus the entrepreneur's share will be less than 60% of the final enterprise value. At very low valuations, the VC's share is 100%. **Figure C1** provides payout diagrams to illustrate these features.

Figure C1 Convertible Preferred Stock Payout Diagrams



Source: Case authors.

The horizontal axis in this figure is the value of the enterprise. The dotted 45 degree line is also the total enterprise value to be divided up between the entrepreneur and the VC at exit. The solid

line is the exit value to the VC. In Zone A, the enterprise value at exit is less than the amount invested by the VC at the last financing round. If the firm's value is only \$2 million, the VC will not convert its \$4 million debt to equity and will receive all \$2 million in value. In this zone, the VC's final share is 100% of the total enterprise value.

In Zone B, the venture's exit value is greater than the VC's investment but less than the post-money valuation from the previous financing round. In this range, it is still not worthwhile for the VC to convert its debt to equity. Even at a \$9 million exit valuation, the value of a 40% equity share (\$3.6 million) is less than the debt value of \$4 million. Throughout this Zone B, the entrepreneur is catching up in terms of exit value and realized share of the exiting venture. It is only when the exit valuation exceeds the post-money valuation of \$10 million that the VC converts into equity. At any higher exit valuation, the entrepreneur and VC split ownership at the equity shares implied in the pre- and post-money formulas.

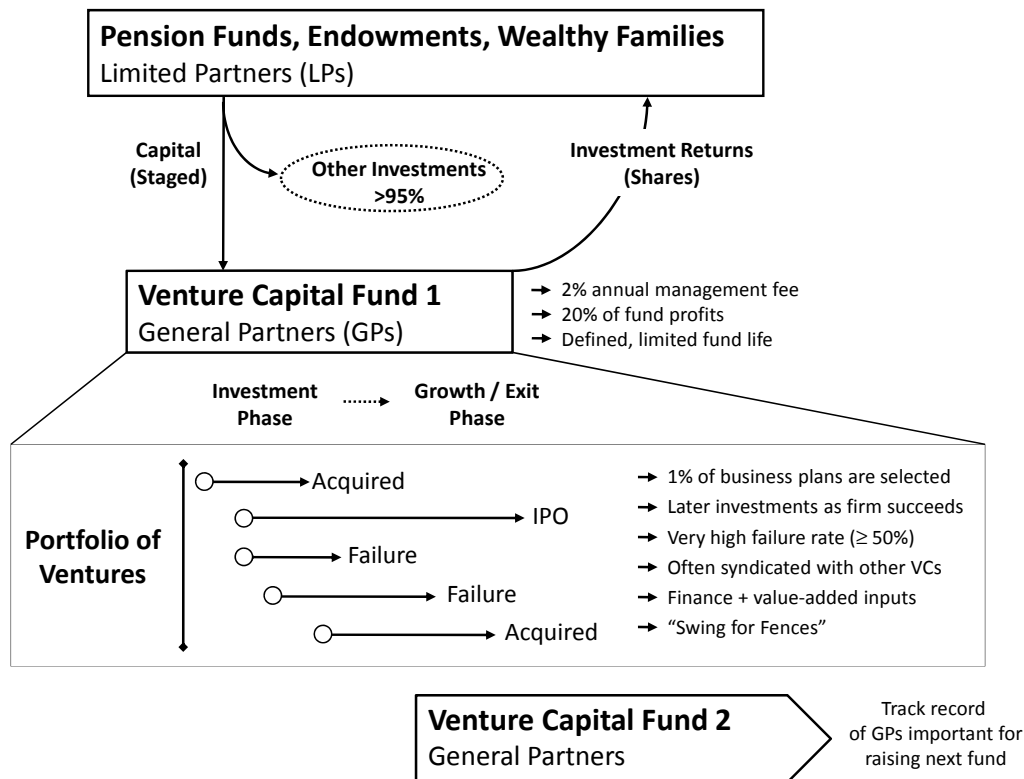
This feature is often poorly understood by entrepreneurs who are focused solely on the very successful outcomes. The reality is that most ventures do not obtain super high exit valuations, and the division of the exit valuation at lower outcomes is very important. More complex instruments like participating preferred stock further secure additional value for the VC firm in moderate exit outcomes (e.g., the participating feature may govern outcomes with exit valuations less than \$50 million). Entrepreneurs need to carefully review and evaluate how these payoffs are structured when negotiating the term sheets provided by VCs.²³

²³ For additional details on participating preferred stocks and other private equity securities, see "A Note of Private Equity Securities," HBS Case No. 200-027 (Boston: Harvard Business School Publishing, 2007), by Felda Hardyman and Josh Lerner.

APPENDIX D: STRUCTURE OF VENTURE CAPITAL FIRMS AND ANGEL GROUPS

This appendix provides a rough sketch of the structure of VC funds and angel investment groups. While it is well beyond our scope to describe these organizations in detail, a short introduction highlights some structural features that are important for entrepreneurs when raising financing from these options.²⁴ **Figure D1** provides a diagram of a typical VC organization.

Figure D1 Structure of a Venture Capital Fund



Source: Case authors.

Starting from the top of the diagram, VCs raise their financial capital for investments from limited partners (LPs). LPs include pension funds, endowments, high net worth individuals, and similar. These investors commit financial capital to the VC as part of an overall investment strategy, including public equity markets, government bonds, etc., with the VC-oriented investments typically being a small share (5% or less, with some legal restrictions in certain cases like pension funds). These capital commitments are staged as the venture fund progresses, and the LPs receive returns from the portfolio as they are realized. These returns are often distributed in the form of shares.

The VC fund itself is managed by general partners (GPs). GPs screen and select ventures, monitor and aid portfolio companies, manage the liquidity events for successful companies, etc. For this, the

²⁴ Resources on the VC model include Josh Lerner and Paul Gompers, *The Venture Capital Cycle* (Cambridge: MIT Press, 1999) and Andrew Metrick and Ayako Yasuda, *Venture Capital & the Finance of Innovation* (New York: John Wiley & Sons, 2007).

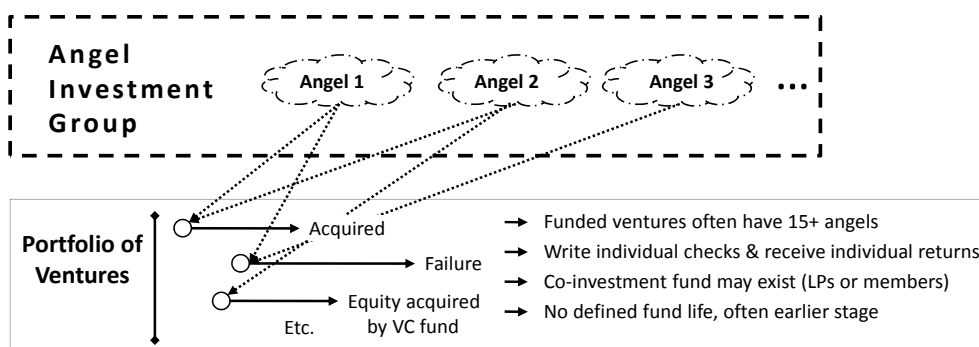
GPs typically receive an annual management fee and a pre-defined share of the fund's profits. A typical arrangement is a 2% management fee and a 20% share of fund profits.

Securities law requires that VC funds have a defined, limited life (often ten years, with the possibilities for short extensions). Thus, VCs raise a series of funds that overlap. Within each fund, the first phase of three to five years involves making investments in companies. The second phase focuses on growth and harvesting the ventures, and investments are not made late in a fund's life as the runway is too short. One feature of this structure is that VCs typically invest with a three to seven year horizon for portfolio companies to realize value.

Finally, there is the portfolio of the investments for the VC fund. This note has highlighted several features of this portfolio. Perhaps the most important conceptual factor is that VCs are seeking home-run opportunities. The VCs know that many of their ventures will fail, but one or two tremendous successes make the economics work. The more that entrepreneurs match the optimal traits for this portfolio (e.g., reasonable capital requirements, moderate time requirements until an exit event is possible, exceptional return potential), the more likely that the VCs will select them. The VCs' success rate in this fund then influences how well the VCs can raise financial capital for the next fund.

Figure D2 shows that the structure of an angel investment group is quite different, although these two financial sources are increasingly overlapping in terms of the entrepreneurial finance landscape. Angel groups are networks of individual angels that meet to evaluate and potentially jointly invest in start-up opportunities. These investment groups provide a variety of advantages to the individual angels: shared due diligence over deals, greater profile to entrepreneurs, opportunity to take on projects requiring larger investments, more extensive networks and backgrounds for portfolio companies, and similar advantages.²⁵

Figure D2 Structure of an Angel Investment Group



Source: Case authors.

While all members of an angel group may meet to evaluate the investments, each angel ultimately makes an individual decision to invest his or her own money. If enough angels are interested, with 15 or 20 often being sufficient critical mass, then investments proceed forward. In some cases, the angel group may have an extra fund with LPs that invests alongside the angels.

²⁵ Resources on angel groups include Lynda Applegate and Kaitlyn Simpson, "CommonAngels A," HBS Case No. 810-082 (Boston: Harvard Business School Publishing, 2010) and William R. Kerr, Josh Lerner, and Antoinette Schoar, "The Consequences of Entrepreneurial Finance: Evidence from Angel Financings", Working Paper (2011).

While angel groups and VCs often have interest in similar deals, their structures and profiles are quite different and this influences investments. First, as angel groups do not have a defined fund life, they may be more interested in earlier stage investment opportunities that take longer to mature. Second, angels invest their own money, in smaller amounts, and perhaps with less diversification. These factors may lead angel groups to invest in opportunities that require less capital and/or are less likely to have an enormous exit valuation (yet still offer high growth and returns). Bigger VC funds need bigger investments and exits for their economics to work, and angel groups often have more flexibility towards smaller investments. These factors may also lead angel groups to focus on management strategies for funded companies that offer moderate returns rather than taking a more risky shot at a very high outcome. These features can limit the degree to which angel groups and VCs co-invest, and entrepreneurs should be cognizant of their basic differences.